CHAPTER -VI

BAND THEORY OF SOLIDS

6.1 INTRODUCTION

The free electron theory described in the previous chapter was successful in explaining the various electronic and thermal properties of metals, such as specific heat, paramagnetism, etc. However, there are various other properties which could not be explained by this theory. For example, the theory could not explain why certain solids have a large number of free electrons and thus behave as good conductors while certain others have hardly any electrons and are, therefore, insulators. It also could not account for the variation of resistivity with temperature for the latter type of materials. Furthermore, the properties of semiconductors could not be explained on the basis of this theory.

The failure of the free electron model is due to the oversimplified assumption that a conduction electron in a metal experiences a constant or zero potential due to the ion cores and hence is free to move about in a crystal; the motion being restrained only by the surface of the crystal. In fact, the potential due to ion cores is not constant and may change with position of the electron in the crystal. Some contribution to potential may also arise because of the other electrons present in the crystal. Thus the actual nature of potential under which an electron moves in a crystal is very complicated. To a reasonable approximation, the ion cores may be considered at rest and the potential experienced by an electron in a crystal is assumed to be periodic with period equal to the lattice constant as shown in Fig. 6.1 for a one-dimensional case. This assumption is based on the fact that the ion cores in a crystal are distributed periodically on the lattice sites. The potential contribution due to all other free electrons may be taken as constant. This type of periodic potential extends up to infinity in all directions except at the surface of the crystal where, due to interrruption in periodicity of the lattice, it has a shape similar to as shown on the right hand edge of the figure. This type of potential irregularity may, however, be ignored.

The periodic potential described above forms the basis of the band theory of solids. The behaviour of an electron in this potential is described by constructing the electron wave functions using one-electron approxima-

tion. In this approximation, the total wave function for the system is obtained from a combination of wave functions each one of which involves the coordinates of one electron. In other words, the field experienced by a particular electron is assumed to be the resultant of the field due to fixed nuclei and the average field due to the charge distribution of all other electrons. As we shall discuss later, the motion of an electron in a periodic potential yields the following results :

(a) There exist allowed energy bands separated by forbidden regions or band gaps.

(b) The electronic energy functions E(**k**) are periodic in the wave vector **k**.

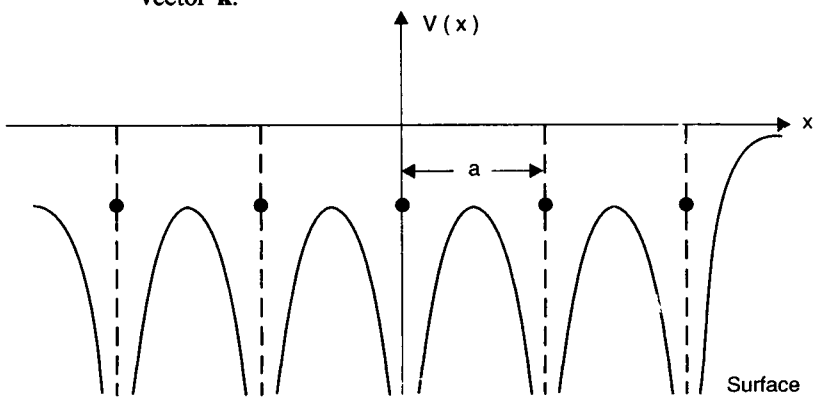


Fig. 6.1. *One-dimensional representation of potential experienced by an electron in a perfectly periodic crystal lattice with lattice constant a. The potential near the surface is shown on the right.*

These results are in contrast with those obtained from the free electron theory (constant or zero potential case) in the sense that, in the free electron theory, E varies with k as

$$E = \frac{\hbar^2 k^2}{2m}$$

i.e., there is no upper limit to the energy and k can have discrete values, which means that the energy levels are discrete and may have any spacing depending on the dimensions of the box. The existence of band gaps is the most important new property which emerges when the free electron model is extended to include the effect of periodic potential of the ion cores. It will be shown that these band gaps result from the interaction of the conduction electron waves with ion cores of the crystal and are of decisive significance to determine whether a solid is a conductor, a semiconductor, or an insulator. Besides this, we shall also encounter some other remarkable properties of electrons in crystals. We first determine the electron wave functions for a periodic lattice.

6.2 THE BLOCH THEOREM

As described in case of the free electron theory, the one-dimensional Schrodinger equation for an electron moving in a constant potential V_o is

$$\frac{d^2\psi}{dx^2} + \frac{2m}{\hbar^2}(E - V_o)\psi = 0 \tag{6.1}$$

The solutions to this equation are plane waves of the type

$$\psi(x) = e^{\pm ikx} \tag{6.2}$$

where

$$(E - V_o) = \frac{\hbar^2 k^2}{2m} = \frac{p^2}{2m} = E_{kin}$$

For an electron moving in a one-dimensional periodic potential, $V(x)$, the Schrodinger equation is written as

$$\frac{d^2\psi}{dx^2} + \frac{2m}{\hbar^2}[E - V(x)]\psi = 0 \tag{6.3}$$

Since the potential is periodic with period equal to the lattice constant, a, we have

$$V(x) = V(x + a)$$

There is an important theorem known as the *Bloch theorem* or the *Floquet's theorem* concerned with the solutions to Eq. (6.3). According to this theorem, the solutions to Eq. (6.3) are plane waves of the type of Eq. (6.2) which are modulated by a function $u_k(x)$ having the same periodicity as that of the lattice. Thus the solutions are of the form

$$\psi(x) = e^{\pm ikx}\, u_k(x) \tag{6.4}$$

with

$$u_k(x) = u_k(x + a) \tag{6.5}$$

The wave functions of the type of Eq. (6.4) are called the *Bloch functions.*

Proof of the Bloch Theorem

Let $g(x)$ and $f(x)$ be two real and independent solutions to the second order differential equation (6.3). Then the general solution can be written as

$$\psi(x) = A\, f(x) + B\, g(x) \tag{6.6}$$

where A and B are arbitrary constants. Since $V(x) = V(x + a)$, the functions $g(x + a)$ and $f(x + a)$ must also be the solutions to Eq. (6.3) in addition to $g(x)$ and $f(x)$

However, Eq. (6.3) cannot have more than two independent solutions, the functions $g(x+a)$ and $f(x+a)$ must be expressible in terms of the functions $g(x)$ and $f(x)$, i.e.,

$$\left.\begin{aligned} f(x+a) &= \alpha_1 f(x) + \alpha_2 g(x) \\ g(x+a) &= \beta_1 f(x) + \beta_2 g(x) \end{aligned}\right] \tag{6.7}$$

where α_1, α_2, β_1 and β_2 are the real functions of energy E. Another general solution to Eq. (6.3) as obtained from the functions (6.7) is

$$\begin{aligned} \psi(x + a) &= A\, f(x + a) + B\, g(x + a) \\ &= (A\alpha_1 + B\beta_1)\, f(x) + (A\alpha_2 + B\beta_2)\, g(x) \end{aligned} \tag{6.8}$$

If the constants A and B are chosen such that

$$\left.\begin{aligned} A\alpha_1 + B\beta_1 &= \lambda A \\ \text{and} \qquad A\alpha_2 + B\beta_2 &= \lambda B \end{aligned}\right] \tag{6.9}$$

where λ is a constant, we find from Eqs. (6.6) and (6.8) that the functions $\psi(x)$ has the property

$$\psi(x + a) = \lambda\psi(x) \tag{6.10}$$

The Eqs. (6.9) yield non-zero values for A and B only if the determinant of the coefficients vanishes, i.e. if

$$\begin{vmatrix} (\alpha_1 - \lambda) & \beta_1 \\ \alpha_2 & (\beta_2 - \lambda) \end{vmatrix} = 0 \tag{6.11}$$

or

$$\lambda^2 - (\alpha_1 + \beta_2)\lambda + \alpha_1\beta_2 - \alpha_2\beta_1 = 0 \tag{6.12}$$

We now show that $\alpha_1\beta_2 - \alpha_2\beta_1 = 1$ Since $g(x)$ and $f(x)$ are the real and independent solutions to Eq. (6.3), we get

$$\frac{d^2g(x)}{dx^2} + \frac{2m}{\hbar^2}\left[E - V(x)\right] g(x) = 0$$

and

$$\frac{d^2f(x)}{dx^2} + \frac{2m}{\hbar^2}\left[E - V(x)\right] f(x) = 0$$

Multiplying the former by $f(x)$ and the latter by $g(x)$ and subtracting, we obtain

$$f(x)\frac{d^2g(x)}{dx^2} - g(x)\frac{d^2f(x)}{dx^2} = 0$$

or $$f(x)\frac{dg(x)}{dx} - g(x)\frac{df(x)}{dx} = \text{constant}$$

The left hand side is called Wronskian, $W(x)$, of the solutions and is constant in this case. Further, from Eqs. (6.7), we obtain

$$f(x+a)\frac{dg(x+a)}{dx} - g(x+a)\frac{df(x+a)}{dx} = (\alpha_1\beta_2 - \alpha_2\beta_1)\left[f(x)\frac{dg(x)}{dx} - g(x)\frac{df(x)}{dx}\right]$$

or $$W(x+a) = (\alpha_1\beta_2 - \alpha_2\beta_1)\ W(x)$$

Hence

$$\alpha_1\beta_2 - \alpha_2\beta_1 = 1 \tag{6.13}$$

Thus, Eq. (6.12) becomes

$$\lambda^2 - (\alpha_1 + \beta_2)\lambda + 1 = 0 \tag{6.14}$$

It gives two possible values of λ and hence two functions $\psi_1(x)$ and $\psi_2(x)$ which have the property as given in Eq. (6.10). We consider the following two cases :

Case I *For energy ranges such that* $(\alpha_1 + \beta_2)^2 < 4$

The Eq. (6.14) will have complex roots λ_1 and λ_2. Since $\lambda_1\lambda_2 = 1$ these will be complex conjugates of each other. We may, therefore, write

$$\left.\begin{aligned} \lambda_1 &= e^{ika} \\ \text{and} \quad \lambda_2 &= e^{-ika} \end{aligned}\right] \tag{6.15}$$

where k is real. The corresponding functions $\psi_1(x)$ and $\psi_2(x)$ then have the property

$$\left.\begin{aligned} \psi_1\ (x + a) &= e^{ika}\ \psi_1(x) \\ \text{and} \quad \psi_2\ (x + a) &= e^{-ika}\ \psi_2(x) \end{aligned}\right] \tag{6.16}$$

or, in general,

$$\psi(x+a) = e^{\pm ika}\ \psi(x) \tag{6.17}$$

It is easy to see that a function having such a property is the Bloch function of the type of Eq. (6.4). To check this we replace x by $x+a$ in the Bloch function (6.4) and obtain

$$\psi(x+a) = e^{\pm ik(x+a)}\ u_k(x+a)$$

Using Eq. (6.5), it becomes

$$\psi(x+a) = e^{\pm ika}\ \psi(x)$$

which is the same as (6.17). Thus we find that the function $\psi(x)$, which represents a solution to Eq. (6.3), has the same property as that expressed by Eq. (6.17), Eq. (6.10) with $\lambda = e^{\pm ika}$, or the Bloch function (6.4). Hence

the Bloch functions of the type of Eq. (6.4) represent the solutions to the Schrodinger equation (6.13). This proves the Bloch theorem.

Case II *For the energy ranges such that* $(\alpha_1 + \beta_2)^2 > 4$

The Eq. (6.14) will have real roots λ_1 and λ_2 which may be taken as

$$\lambda_1 = e^{\mu a} \quad \text{and} \quad \lambda_2 = e^{-\mu a} \tag{6.18}$$

where μ is real. The corresponding solutions to the Schrodinger equation are

$$\psi_1(x) = e^{\mu x} u(x) \quad \text{and} \quad \psi_2(x) = e^{-\mu x}\, u(x) \tag{6.19}$$

Though mathematically valid, these solutions are not the acceptable electron wave functions because these are not bounded. The ratio $\dfrac{\psi(x+na)}{\psi(x)}$ approaches infinity as n approaches $\pm\infty$. This is contradictory to the requirement that the wave functions should remain finite at all points in space because of their probabilistic interpretation. In case I, however, $|\psi(x+na)|$ is equal to $|\psi(x)|$ and the ratio of these is always finite.

The roots (6.15) and (6.18) depend on α and β and hence are functions of energy E. The allowed roots (6.15) are associated with allowed energy regions and the disallowed roots (6.18) are associated with the forbidden energy regions. This means that the energy spectrum of an electron moving in a periodic potential consists of allowed and forbidden energy regions or bands. This result will be illustrated using the Kronig-Penny model.

In three dimensions, the Bloch theorem is expressed as

$$\psi_k(\mathbf{r}) = e^{i\mathbf{k}.\mathbf{r}}\, u_k(\mathbf{r}) \tag{6.20}$$

If the lattice potential vanishes, $u_k(\mathbf{r})$ becomes constant and Eq. (6.20) gives

$$\psi_k(\mathbf{r}) = e^{i\mathbf{k}.\mathbf{r}}$$

Thus the wave function becomes the one of a free electron.

6.3 THE KRONIG-PENNEY MODEL

This model illustrates the behaviour of electrons in a periodic potential by assuming a relatively simple one-dimensional model of periodic potential as shown in Fig. 6.2. The potential energy of an electron in a linear array of positive nuclei is assumed to have the form of a periodic array of square wells with period of $(a + b)$. At the bottom of a well, i.e. for $0 < x < a$, the electron is assumed to be in the vicinity of a nucleus and the potential energy is taken as zero, whereas outside a well, i.e. for $-b < x < 0$, the potential energy is assumed to be V_0. Although this model employs a very crude approximation to the type of periodic potential existing inside a lattice, yet it is very useful to illustrate various important features of the quantum behaviour of electrons in the periodic lattice. The wave functions

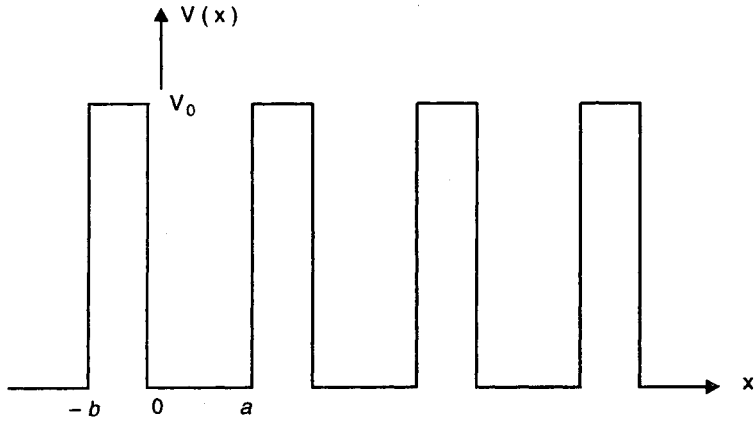


Fig. 6.2. *One-dimensional periodic potential used by Kronig and Penney.*

are obtained by writing the Schrodinger equations for the two regions as

$$\frac{d^2\psi}{dx^2} + \frac{2m}{\hbar^2} E\psi = 0 \qquad \text{for } 0 < x < a \tag{6.21}$$

and

$$\frac{d^2\psi}{dx^2} + \frac{2m}{\hbar^2} (E - V_0)\psi = 0 \qquad \text{for } -b < x < 0 \tag{6.22}$$

Assuming that the energy E of the electrons is less than V_0, we define two real quantities α and β as

$$\alpha^2 = \frac{2mE}{\hbar^2} \quad \text{and} \quad \beta^2 = \frac{2m(V_0 - E)}{\hbar^2} \tag{6.23}$$

Therefore, the Eqs. (6.21) and (6.22) become

$$\frac{d^2\psi}{dx^2} + \alpha^2\psi = 0 \qquad \text{for } 0 < x < a \tag{6.24}$$

and

$$\frac{d^2\psi}{dx^2} - \beta^2\psi = 0 \qquad \text{for } -b < x < 0 \tag{6.25}$$

Since the potential is periodic, the wave functions must be of the form of Bloch function (6.4), i.e.,

$$\psi(x) = e^{ikx}\, u_k(x) \tag{6.26}$$

where $u_k(x)$ is the periodic function in x with periodicity of $(a+b)$, i.e.,

$$u_k(x) = u_k(x + a + b) \tag{6.27}$$

From Eq. (6.26), we have

$$\frac{d\psi}{dx} = ike^{ikx}u_k(x) + e^{ikx}\frac{du_k}{dx}$$

and

$$\frac{d^2\psi}{dx^2} = -k^2e^{ikx}u_k(x) + 2ike^{ikx}\frac{du_k}{dx} + e^{ikx}\frac{d^2u_k}{dx^2} \tag{6.28}$$

Substituting Eqs. (6.26) and (6.28) in Eqs. (6.24) and (6.25), we get

$$\frac{d^2u_1}{dx^2} + 2ik\frac{du_1}{dx} + \left(\alpha^2 - k^2\right)u_1 = 0 \qquad 0 < x < a \tag{6.29}$$

and

$$\frac{d^2u_2}{dx^2} + 2ik\frac{du_2}{dx} - \left(\beta^2 + k^2\right)u_2 = 0; \qquad -b < x < 0 \tag{6.30}$$

where u_1 and u_2 represent the values of $u_k(x)$ in the intervals $0 < x < a$. and $-b < x < 0$ respectively. The general solutions to these equations are

$$u_1 = A\,e^{i(\alpha-k)x} + B\,e^{-i(\alpha+k)x}; \qquad 0 < x < a \tag{6.31}$$

and

$$u_2 = C\,e^{(\beta-ik)x} + D\,e^{-(\beta+ik)x}; \qquad -b < x < 0 \tag{6.32}$$

where A, B, C and D are constants which can be determined from the following boundary conditions :

$$u_1(0) = u_2(0); \qquad \left(\frac{du_1}{dx}\right)_{x=0} = \left(\frac{du_2}{dx}\right)_{x=0} \tag{6.33}$$

$$u_1(a) = u_2(-b); \qquad \left(\frac{du_1}{dx}\right)_{x=a} = \left(\frac{du_2}{dx}\right)_{x=-b} \tag{6.34}$$

The conditions (6.33) correspond to the requirements that the wave function ψ and its derivative $\frac{d\psi}{dx}$, and hence u and $\frac{du}{dx}$ must be continuous, whereas the conditions (6.34) correspond to the requirements of the periodicity of $u_k(x)$. Applying these boundary conditions to (6.31) and (6.32) we obtain the following four linear homogeneous equations :

$$\begin{aligned}
A + B &= C + D \\
Ai(\alpha-k) - Bi(\alpha+k) &= C(\beta-ik) - D(\beta+ik) \\
Ae^{i(\alpha-k)a} + Be^{-i(\alpha+k)a} &= Ce^{-(\beta-ik)b} + De^{(\beta+ik)b} \\
Ai(\alpha-k)e^{i(\alpha-k)a} - Bi(\alpha+k)e^{-i(\alpha+k)a} &= C(\beta-ik)e^{-(\beta-ik)b} - Di(\beta+ik)e^{(\beta+ik)b}
\end{aligned} \tag{6.35}$$

These equations are used to determine the constants A, B, C and D. A non-zero solution to these equations exists only if the determinant of the coefficients of A, B, C and D vanishes, i.e.,

$$\begin{vmatrix} 1 & 1 & 1 & 1 \\ i(\alpha-k) & -i(\alpha+k) & \beta-ik & -(\beta+ik) \\ e^{i(\alpha-k)a} & e^{-i(\alpha+k)a} & e^{-(\beta-ik)b} & e^{(\beta+ik)b} \\ i(\alpha-k)e^{i(\alpha-k)a} & -i(\alpha+k)e^{-i(\alpha+k)a} & (\beta-ik)e^{-(\beta-ik)b} & -(\beta+ik)e^{(\beta+ik)b} \end{vmatrix} = 0$$

On solving this determinant, we obtain

$$\frac{\beta^2+\alpha^2}{2\beta\alpha}\sinh\beta b \sin\alpha a + \cosh\beta b \cos\alpha a = \cos k(a+b) \tag{6.36}$$

To simplify this equation, Kronig and Penney considered the case when V_0 tends to infinity and b approaches zero but the product $V_0 b$ remains finite, i.e., the potential barriers become delta functions. Under these circumstances, the model is modified to the one of a series of wells separated by infinitely thin potential barriers. The quantity $\lim V_0 b$ for $V_0 \to \infty$ and $b \to 0$ is called the *barrier strength.*

As $b \to 0$, $\sinh \beta b \to \beta b$ and $\cosh \beta b \to 1$. Also, from Eqs. (6.23), we have

$$\frac{\beta^2+\alpha^2}{2\alpha\beta} = \frac{mV_0}{\alpha\beta\hbar^2}$$

Therefore, Eq. (6.36) becomes

$$\left(\frac{mV_0 b}{\hbar^2\alpha}\right)\sin\alpha a + \cos\alpha a = \cos ka \tag{6.37}$$

We define a quantity P as

$$P = \frac{mV_0 ba}{\hbar^2} \tag{6.38}$$

which is a measure of the area $V_0 b$ of the potential barrier. Thus increasing P has the physical meaning of binding an electron more strongly to a particular potential well. Using Eq. (6.38) in (6.37), we get

$$P\frac{\sin\alpha a}{\alpha a} + \cos\alpha a = \cos ka \tag{6.39}$$

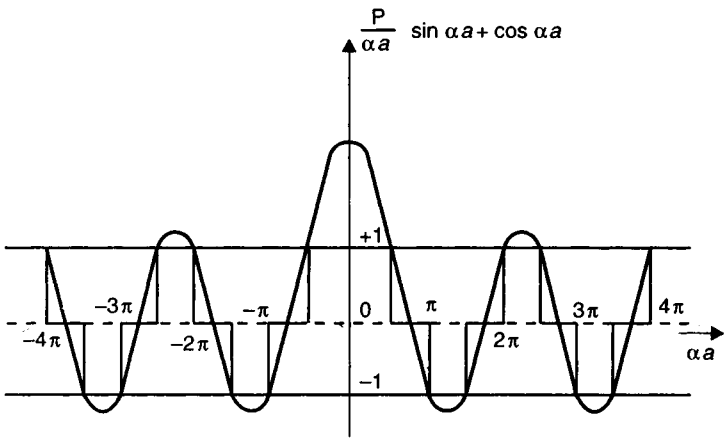


Fig. 6.3. *Plot of the left hand side of Eq. (6.39) as a function of αa using P = 3π/2. The solid and broken lines on the abscissa correspond to allowed and forbidden energy regions respectively.*

This is the condition which must be satisfied for the solutions to the wave equation to exit. Since cos*ka* lies between +1 and −1, the left hand side should assume only those values of αa for which its value lies between +1 and −1. Such values of αa, therefore, represent wave like solutions of the type of Eq. (6.26) and are allowed. The other values of αa are not allowed. Figure 6.3 shows a plot of the left hand side versus αa for $P = 3\pi/2$. The vertical axis lying between −1 and +1, as indicated by the horizontal lines, represents the values acceptable to the left hand side. It may be noted that since α^2 is proportional to the energy E, the abscissa is a measure of the energy. The following conclusions may be drawn from Fig. 6.3 :

(i) The energy spectrum of the electrons consists of alternate regions of allowed energy bands (solid lines on abscissa) and forbidden energy bands (broken lines).

(ii) The width of the allowed energy bands increases with αa or the energy.

(iii) The width of a particular allowed energy band decreases with increase in the value of P, i.e., with increase in the binding energy of the electrons. As $P \to \infty$, the allowed energy bands are compressed into energy levels and a line spectrum is resulted. In such a case, the Eq. (6.39) has solutions only if

$$\sin\alpha a = 0$$

or $$\alpha a = \pm n\pi$$

where n is an integer. Therefore, from Eq. (6.23), we have

$$E = \frac{\alpha^2\hbar^2}{2m} = \frac{\pi^2\hbar^2}{2ma^2}n^2 \tag{6.40}$$

This expression gives the energy levels of a particle in a constant potential box of atomic dimensions. This is what is expected physically as, for large P, the tunnelling through the barrier becomes difficult. In the other extreme case, i.e., when $P \to 0$, Eq. (6.39) becomes

$$\cos \alpha a = \cos ka$$

or $$\alpha = k$$

Therefore, Eq. (6.23) gives

$$E = \frac{k^2\hbar^2}{2m}$$

which is applicable to completely free electrons for which all the energies are allowed. This is again an expected result as, for P approaching zero, the electrons become free and, therefore, the free electron model involving quasi-continuous energy levels becomes applicable.

These results are summarized in Fig. 6.4 which shows the energy spectrum as a function of P. On the extreme left, i.e. for $P = 0$, the energy spectrum is quasi-continuous and on the extreme right, i.e. for $P = \infty$, a line spectrum is resulted. For any other value of P, the position and width of the allowed and forbidden bands are obtained by drawing a vertical line.

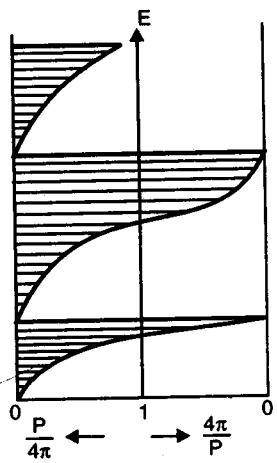


Fig. 6.4. *Allowed (shaded) and forbidden (open) energy ranges as a function of P.*

6.3.1 Energy versus Wave-Vector Relationship — Different Representations

It follows from Eq. (6.39) that coska takes a specific value for each allowed energy value E. However, coska is an even periodic function of k and remains unchanged if k is replaced by $k + 2\pi n/a$, where n is an integer. This means that the energy E is also an even periodic function of k with period of $2\pi/a$. This type of periodic representation of energy is shown in Fig. 6.5a for the first few allowed bands. This representation can be obtained by the periodic repetition of the region $-\pi/a < k < \pi/a$, i.e., the first *Brillouin*

zone, and is known as the *periodic or repeated zone scheme.* In this scheme, k is not uniquely defined as, for each value of E, there exist a number of k values. It is, therefore, generally convenient to introduce the following two schemes :

(*a*) The extended zone scheme

(*b*) The reduced zone scheme

A curve of E versus k in the extended zone scheme is shown by solid lines in Fig. 6.5b. The corresponding parabolic curve (dotted) for free electrons in the constant potential is also shown for comparison. We note that the E–k curve of the extended zone scheme is not continuous and has discontinuities at

$$k = \pm \frac{n\pi}{a} \tag{6.41}$$

where $n = 1, 2, 3, \ldots$ These values of k define the boundaries of the Brillouin zones. As described in Chapter II, the first Brillouin zone extends from $-\pi/a$ to $+\pi/a$, the second one extends from π/a to $2\pi/a$ and from $-\pi/a$ to $2\pi/a$, and so on. Since $k = 2\pi/\lambda$, Eq. (6.41) gives

$$n\lambda = 2a \tag{6.42}$$

which is the Bragg's law of reflection for normal incidence. It thus follows that, for the values of k as given by (6.41), the electrons suffer Bragg reflection which results in discontinuities in the E–k curve.

The E–k curve in the reduced zone scheme is shown in Fig. 6.5c It is obtained by 'reducing' the contents of the other zones so as to correspond, in general, to the first zone, i.e., to the region $-\pi/a \leq k \leq \pi/a$. The wave vector **k** belonging to this region is called the *reduced wave vector* and can be obtained from the wave vector **k** belonging to the region outside the first Brillouin zone by using the relation

$$\mathbf{k} = \mathbf{k}' - \mathbf{G} \tag{6.43}$$

where **G** is a suitably chosen reciprocal lattice vector.

6.3.2 Number of Wave Functions in a Band

It has been shown that, in an infinitely long one-dimensional crystal, there are certain allowed energy ranges which have a continuous distribution of energy. We now consider a finite crystal of length L and determine the number of possible wave functions, states or k-values per band. Using the periodic boundary conditions, we have

$$\psi(x + L) = \psi(x) \tag{6.44}$$

Since the wave functions are Bloch functions, it requires